Lesson 2: Topology and Continuity

AI/ML Mathematics

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1 Motivation and Overview

Consider training a neural network $f_\theta \colon \mathbb{R}^d \to \mathbb{R}$ to minimize a loss function $L(\theta)$ over a parameter space $\Theta \subseteq \mathbb{R}^d$. A natural question is: does the minimum of L over Θ actually exist, or could the infimum be a value that is never attained? In finite dimensions, the *extreme value theorem* guarantees that a continuous function on a *compact* set attains its maximum and minimum. But what exactly is compactness, and why does it work? Understanding this requires the topology of the real line—open sets, closed sets, and the Heine–Borel characterization of compact subsets of $\mathbb{R}^n$—together with the precise ε-δ definition of continuity. The tools developed in this lesson will let us prove that continuous functions on closed bounded sets attain their extreme values, that the intermediate value theorem provides existence of roots and fixed points, and that uniform continuity gives the quantitative control needed for approximation arguments throughout optimization and reinforcement learning.

Why RL/ML needs topology and continuity

- **Existence of optimal solutions.** The extreme value theorem guarantees that a continuous loss function attains its minimum on a compact parameter set—a foundational assumption in optimization.
- **Value function continuity.** In MDPs with continuous state spaces, the value function V^π is often shown to be continuous; this enables function approximation and ensures nearby states have similar values.
- **Compactness in function approximation.** The Arzelà–Ascoli theorem (a compactness criterion for function spaces) underpins universal approximation results for neural networks.
- **Fixed-point existence.** Brouwer's fixed-point theorem (every continuous map from a compact convex set to itself has a fixed point) is used to prove existence of Nash equilibria and optimal policies.

Prerequisites. Completeness of $\mathbb{R}$ (the least upper bound property), convergence of sequences (the ε-N definition), the algebraic limit theorems, the Bolzano–Weierstrass theorem (Lesson 1, Theorem 4.7), and the Cauchy criterion for convergence (Lesson 1, Theorem 4.10).

Learning objectives. After completing this lesson, you will be able to:

- **Define** open sets, closed sets, closure, interior, and boundary in $\mathbb{R}$, and **classify** sets using these concepts.
- **State** and **prove** the Heine–Borel theorem characterizing compact subsets of $\mathbb{R}$.
- **Prove** that sequential compactness and open-cover compactness are equivalent in $\mathbb{R}$.
- **Define** continuity via ε-δ and via sequences, and **prove** their equivalence.
- **Derive** the extreme value theorem and the intermediate value theorem from compactness and continuity.
- **Prove** that a continuous function on a compact set is uniformly continuous.

2 References

- Rudin [1976, Chapters 2 and 4] — Topology of the real line, compactness, and continuity with complete proofs.
- Abbott [2015, Chapters 3–4] — An accessible treatment of open and closed sets, compact sets, and the extreme value theorem.

- Bartle and Sherbert [2011, Chapter 4] — Detailed development of continuity with many worked examples and exercises.
- Pugh [2015, Chapters 2–3] — A geometric perspective on topology and continuity in $\mathbb{R}$ and $\mathbb{R}^n$.
- Folland [1999, Chapter 1] — Point-set topology foundations as preparation for measure theory.

3 Open and Closed Sets

Definition 3.1 (ε-Neighborhood)**.** For $x \in \mathbb{R}$ and $\varepsilon > 0$, the *ε-neighborhood* of x is the open interval $V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |y - x| < \varepsilon\}$.

Definition 3.2 (Open Set)**.** A set $O \subseteq \mathbb{R}$ is *open* if for every $x \in O$ there exists $\varepsilon > 0$ such that $V_\varepsilon(x) \subseteq O$.

Example 3.3. Every open interval (a, b) is an open set. If $x \in (a, b)$, take $\varepsilon = \min(x-a, b-x) > 0$. Then $V_\varepsilon(x) \subseteq (a, b)$.

Example 3.4. The entire real line $\mathbb{R}$ is open (take any $\varepsilon > 0$), and the empty set $\emptyset$ is open (the condition holds vacuously).

Theorem 3.5 (Properties of Open Sets)**.** *(i) The union of any collection of open sets is open.*
(ii) The intersection of finitely many open sets is open.

Proof. (i) Let $\{O_\alpha\}_{\alpha \in A}$ be a collection of open sets and $O = \bigcup_{\alpha \in A} O_\alpha$. If $x \in O$, then $x \in O_\beta$ for some $\beta \in A$. Since O_β is open, there exists $\varepsilon > 0$ with $V_\varepsilon(x) \subseteq O_\beta \subseteq O$.

(ii) Let $O_1, \ldots, O_n$ be open and $x \in O_1 \cap \cdots \cap O_n$. For each k, there exists $\varepsilon_k > 0$ with $V_{\varepsilon_k}(x) \subseteq O_k$. Set $\varepsilon = \min(\varepsilon_1, \ldots, \varepsilon_n) > 0$. Then $V_\varepsilon(x) \subseteq O_k$ for all k, so $V_\varepsilon(x) \subseteq O_1 \cap \cdots \cap O_n$. □

Infinite intersections of open sets need not be open

The finite intersection condition in Theorem 3.5(ii) cannot be extended to infinite intersections. For example, $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$, which is not open.

Definition 3.6 (Closed Set)**.** A set $F \subseteq \mathbb{R}$ is *closed* if its complement $F^c = \mathbb{R} \setminus F$ is open.

Example 3.7. The closed interval $[a, b]$ is closed: its complement $(-\infty, a) \cup (b, \infty)$ is a union of open sets, hence open by Theorem 3.5(i).

Theorem 3.8 (Sequential Characterization of Closed Sets)**.** *A set $F \subseteq \mathbb{R}$ is closed if and only if every convergent sequence (x_n) with $x_n \in F$ for all n has its limit in F.*

Proof. ($\Rightarrow$) Suppose F is closed and $x_n \to L$ with $x_n \in F$. Assume for contradiction that $L \notin F$. Then $L \in F^c$, which is open, so there exists $\varepsilon > 0$ with $V_\varepsilon(L) \subseteq F^c$. But $x_n \to L$ means there exists N with $|x_n - L| < \varepsilon$ for $n \geq N$, giving $x_N \in F^c$, a contradiction since $x_N \in F$.

($\Leftarrow$) Suppose F satisfies the sequential condition. We show F^c is open. Let $x \in F^c$. If no ε-neighborhood of x is contained in F^c, then for each $n \in \mathbb{N}$ there exists $x_n \in F \cap V_{1/n}(x)$. Then $x_n \to x$ with $x_n \in F$, so $x \in F$ by hypothesis, contradicting $x \in F^c$. Hence some $V_\varepsilon(x) \subseteq F^c$, and F^c is open. □

Definition 3.9 (Interior, Closure, Boundary)**.** Let $A \subseteq \mathbb{R}$.

(i) The *interior* of A is $\operatorname{int}(A) = \{x \in A : \exists \varepsilon > 0,\ V_\varepsilon(x) \subseteq A\}$. It is the largest open set contained in A.

(ii) The *closure* of A is $\mathrm{cl}(A) = \{x \in \mathbb{R} : \text{every } V_\varepsilon(x) \text{ intersects } A\}$. Equivalently, $\mathrm{cl}(A)$ is the smallest closed set containing A.
(iii) The *boundary* of A is $\partial A = \mathrm{cl}(A) \setminus \mathrm{int}(A)$.

Proposition 3.10 (Sequential Characterization of Closure)**.** *$x \in \mathrm{cl}(A)$ if and only if there exists a sequence (a_n) in A with $a_n \to x$.*

Proof. ($\Rightarrow$) If $x \in \mathrm{cl}(A)$, then for each $n \in \mathbb{N}$, $V_{1/n}(x) \cap A \neq \emptyset$, so we can choose $a_n \in V_{1/n}(x) \cap A$. Then $|a_n - x| < 1/n$, so $a_n \to x$.

($\Leftarrow$) If $a_n \to x$ with $a_n \in A$, then for any $\varepsilon > 0$, there exists N with $a_N \in V_\varepsilon(x) \cap A$. Hence $V_\varepsilon(x) \cap A \neq \emptyset$ for every $\varepsilon > 0$, so $x \in \mathrm{cl}(A)$. □

Example 3.11. The closure $\mathrm{cl}(\mathbb{Q}) = \mathbb{R}$, since every real number is the limit of a sequence of rationals (by the density of $\mathbb{Q}$ in $\mathbb{R}$, Lesson 1, Theorem 3.11). Similarly, $\mathrm{int}(\mathbb{Q}) = \emptyset$, since no open interval is contained in $\mathbb{Q}$ (every interval contains irrationals). Hence $\partial\mathbb{Q} = \mathbb{R}$.

With the basic notions of open and closed sets in place, we now turn to compactness, the property that makes topology most useful for analysis.

4 Compactness

Definition 4.1 (Open Cover, Compactness)**.** Let $K \subseteq \mathbb{R}$.
(i) An *open cover* of K is a collection $\{O_\alpha\}_{\alpha \in A}$ of open sets whose union contains K: $K \subseteq \bigcup_{\alpha \in A} O_\alpha$.
(ii) A *finite subcover* is a finite subcollection $\{O_{\alpha_1}, \ldots, O_{\alpha_n}\}$ that still covers K.
(iii) K is *compact* if every open cover of K has a finite subcover.

Definition 4.2 (Sequential Compactness)**.** A set $K \subseteq \mathbb{R}$ is *sequentially compact* if every sequence (x_n) in K has a subsequence that converges to a point in K.

Theorem 4.3 (Characterization of Compact Sets in $\mathbb{R}$)**.** *For $K \subseteq \mathbb{R}$, the following are equivalent:*
(i) K is compact (every open cover has a finite subcover).
(ii) K is sequentially compact.
(iii) K is closed and bounded.

We prove this theorem through a sequence of implications.

Lemma 4.4. *If K is compact, then K is closed and bounded.*

Proof. Bounded: The collection $\{(-n, n) : n \in \mathbb{N}\}$ is an open cover of K (since $\mathbb{R} = \bigcup_{n=1}^{\infty}(-n, n)$). By compactness, finitely many suffice: $K \subseteq (-n_1, n_1) \cup \cdots \cup (-n_m, n_m) = (-N, N)$ where $N = \max(n_1, \ldots, n_m)$. Hence K is bounded.

Closed: We show K^c is open. Let $y \in K^c$. For each $x \in K$, let $\varepsilon_x = |x - y|/2 > 0$ (since $y \neq x$). The open intervals $V_{\varepsilon_x}(x)$ cover K. By compactness, finitely many suffice: $K \subseteq V_{\varepsilon_{x_1}}(x_1) \cup \cdots \cup V_{\varepsilon_{x_m}}(x_m)$. Set $\delta = \min(\varepsilon_{x_1}, \ldots, \varepsilon_{x_m}) > 0$. Then $V_\delta(y) \cap V_{\varepsilon_{x_k}}(x_k) = \emptyset$ for each k (since $|y - x_k| = 2\varepsilon_{x_k} \geq 2\delta$, and if z were in both then $|x_k - y| \leq |x_k - z| + |z - y| < \varepsilon_{x_k} + \delta \leq 2\varepsilon_{x_k} = |x_k - y|$, a contradiction). Hence $V_\delta(y) \subseteq K^c$, so K^c is open. □

Lemma 4.5. *If K is closed and bounded, then K is sequentially compact.*

Proof. Let (x_n) be a sequence in K. Since K is bounded, (x_n) is bounded. By the Bolzano–Weierstrass theorem (Lesson 1, Theorem 4.7), (x_n) has a convergent subsequence $x_{n_k} \to L$. Since K is closed, $L \in K$ by Theorem 3.8. Hence K is sequentially compact. □

Lemma 4.6. *If K is sequentially compact, then K is compact.*

Proof. Let $\{O_\alpha\}_{\alpha\in A}$ be an open cover of K. We proceed in two steps.

Step 1 (Lebesgue number). We claim there exists $\delta > 0$ such that for every $x \in K$, the interval $(x-\delta, x+\delta)$ is contained in some O_α. Suppose not: for each $n \in \mathbb{N}$, there exists $x_n \in K$ such that $(x_n - 1/n, x_n + 1/n)$ is not contained in any single O_α. By sequential compactness, (x_n) has a subsequence $x_{n_k} \to L \in K$. Since $L \in K$ and $\{O_\alpha\}$ covers K, there exists $\beta \in A$ with $L \in O_\beta$. Since O_β is open, there exists $\varepsilon > 0$ with $(L-\varepsilon, L+\varepsilon) \subseteq O_\beta$. Choose k large enough that $|x_{n_k} - L| < \varepsilon/2$ and $1/n_k < \varepsilon/2$. Then $(x_{n_k} - 1/n_k, x_{n_k} + 1/n_k) \subseteq (L-\varepsilon, L+\varepsilon) \subseteq O_\beta$, contradicting the choice of x_{n_k}.

Step 2 (Finite subcover). Since K is sequentially compact, it is bounded (otherwise, choosing $x_n \in K$ with $|x_n| > n$ would give a sequence with no convergent subsequence). Say $K \subseteq [-M, M]$. Partition $[-M, M]$ into subintervals of length less than δ; this requires at most $N = \lceil 2M/\delta \rceil$ subintervals. Each subinterval that intersects K has diameter less than δ, so it fits inside some O_α (by Step 1, applied to any point of K in that subinterval). Selecting one such O_α for each of the at most N subintervals gives a finite subcover. □

Proof of Theorem 4.3. Lemma 4.4 gives (i)⇒(iii), Lemma 4.5 gives (iii)⇒(ii), and Lemma 4.6 gives (ii)⇒(i). □

Heine–Borel Theorem

Corollary 4.7 (Heine–Borel)**.** *A subset of $\mathbb{R}$ is compact if and only if it is closed and bounded.*

This is the equivalence (i)⇔(iii) of Theorem 4.3. In $\mathbb{R}^n$, the same characterization holds: $K \subseteq \mathbb{R}^n$ is compact if and only if K is closed and bounded.

Compactness in optimization and function approximation

Compactness ensures that optimization problems have solutions. When we write $\min_{\theta\in\Theta} L(\theta)$ with Θ compact and L continuous, the extreme value theorem (proved below) guarantees the minimum is attained. In RL, compact action spaces ensure that $\max_a Q(s,a)$ is achieved; without compactness, the supremum might not be attained, and greedy policies might not exist. In function approximation, the Arzelà–Ascoli theorem uses compactness (of the domain and of function families) to extract uniformly convergent subsequences from bounded equicontinuous families.

Example 4.8. (a) $[0,1]$ is compact (closed and bounded).
(b) $(0,1)$ is not compact (not closed: $1/n \to 0 \notin (0,1)$).
(c) $[0,\infty)$ is not compact (not bounded).
(d) $\{1/n : n \in \mathbb{N}\} \cup \{0\}$ is compact (closed and bounded).
(e) $\{1/n : n \in \mathbb{N}\}$ is not compact (not closed: $1/n \to 0$ but 0 is not in the set).

Proposition 4.9 (Closed Subsets of Compact Sets)**.** *Every closed subset of a compact set is compact.*

Proof. Let $F \subseteq K$ with F closed and K compact. Then K is closed and bounded (Heine–Borel). Since $F \subseteq K$, F is bounded. Since F is closed and bounded, F is compact by Heine–Borel. □

With compactness established, we now develop continuity and show how these two concepts interact to produce the central theorems of analysis.

5 Continuity

Definition 5.1 (Continuity at a Point)**.** Let $f\colon A \to \mathbb{R}$ with $A \subseteq \mathbb{R}$, and let $c \in A$. Then f is *continuous at* c if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x) - f(c)| < \varepsilon$ whenever $x \in A$ and $|x - c| < \delta$. We say f is *continuous on* A if it is continuous at every point of A.

Theorem 5.2 (Sequential Characterization of Continuity)**.** *Let $f\colon A \to \mathbb{R}$ and $c \in A$. Then f is continuous at c if and only if for every sequence (x_n) in A with $x_n \to c$, we have $f(x_n) \to f(c)$.*

Proof. ($\Rightarrow$) Suppose f is continuous at c and $x_n \to c$. Given $\varepsilon > 0$, choose $\delta > 0$ from the definition of continuity. Since $x_n \to c$, there exists N with $|x_n - c| < \delta$ for $n \geq N$. Then $|f(x_n) - f(c)| < \varepsilon$ for $n \geq N$, so $f(x_n) \to f(c)$.

($\Leftarrow$) Suppose f is not continuous at c. Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ there exists $x \in A$ with $|x - c| < \delta$ but $|f(x) - f(c)| \geq \varepsilon_0$. Taking $\delta = 1/n$ for each $n \in \mathbb{N}$ yields a sequence (x_n) in A with $x_n \to c$ but $|f(x_n) - f(c)| \geq \varepsilon_0$ for all n, so $f(x_n) \not\to f(c)$. $\square$

Example 5.3. Every polynomial $p(x) = a_0 + a_1 x + \cdots + a_d x^d$ is continuous on $\mathbb{R}$. By the algebraic limit theorems (Lesson 1, Theorem 4.4), if $x_n \to c$ then $x_n^k \to c^k$ for each k (by induction on k, using the product rule), and $p(x_n) = a_0 + a_1 x_n + \cdots + a_d x_n^d \to p(c)$ (by the sum rule). The sequential characterization (Theorem 5.2) then gives continuity.

Theorem 5.4 (Algebraic Properties of Continuous Functions)**.** *Let $f, g\colon A \to \mathbb{R}$ be continuous at $c \in A$. Then:*

(i) $f + g$ is continuous at c.
(ii) $f \cdot g$ is continuous at c.
(iii) f/g is continuous at c, provided $g(c) \neq 0$.

Proof. All three follow immediately from Theorem 5.2 and the algebraic limit theorems (Lesson 1, Theorem 4.4): if $x_n \to c$, then $f(x_n) \to f(c)$ and $g(x_n) \to g(c)$, so $(f + g)(x_n) = f(x_n) + g(x_n) \to f(c) + g(c)$, and similarly for products and quotients. $\square$

Theorem 5.5 (Composition of Continuous Functions)**.** *Let $f\colon A \to \mathbb{R}$ be continuous at c, and let $g\colon B \to \mathbb{R}$ be continuous at $f(c)$, where $f(A) \subseteq B$. Then $g \circ f$ is continuous at c.*

Proof. Let $x_n \to c$ with $x_n \in A$. Continuity of f at c gives $f(x_n) \to f(c)$. Since $f(x_n) \in B$ and g is continuous at $f(c)$, we get $g(f(x_n)) \to g(f(c))$. By the sequential characterization, $g \circ f$ is continuous at c. $\square$

Example 5.6. The function $h(x) = \sqrt{1 + x^2}$ is continuous on $\mathbb{R}$: it is the composition of $f(x) = 1 + x^2$ (continuous, being a polynomial) and $g(y) = \sqrt{y}$ (continuous on $[0, \infty)$).

Theorem 5.7 (Preservation of Closed and Open Sets)**.** *Let $f\colon A \to \mathbb{R}$ be continuous on A.*

(i) If $O \subseteq \mathbb{R}$ is open, then $f^{-1}(O) = \{x \in A : f(x) \in O\}$ is open relative to A.
(ii) If $F \subseteq \mathbb{R}$ is closed, then $f^{-1}(F)$ is closed relative to A.

Proof. (i) Let $c \in f^{-1}(O)$, so $f(c) \in O$. Since O is open, there exists $\varepsilon > 0$ with $(f(c) - \varepsilon, f(c) + \varepsilon) \subseteq O$. By continuity, there exists $\delta > 0$ with $|f(x) - f(c)| < \varepsilon$ whenever $x \in A$ and $|x - c| < \delta$. Hence $\{x \in A : |x - c| < \delta\} \subseteq f^{-1}(O)$.

(ii) $f^{-1}(F) = A \setminus f^{-1}(F^c)$, and F^c is open, so $f^{-1}(F^c)$ is open relative to A by (i). Hence $f^{-1}(F)$ is closed relative to A. $\square$

We now combine continuity with compactness to obtain the most powerful results of this lesson.

6 Extreme Value Theorem and Intermediate Value Theorem

Theorem 6.1 (Image of Compact Set is Compact)**.** *Let $f\colon K \to \mathbb{R}$ be continuous and K compact. Then $f(K)$ is compact.*

Proof. We use sequential compactness. Let (y_n) be a sequence in $f(K)$. For each n, choose $x_n \in K$ with $f(x_n) = y_n$. Since K is compact (hence sequentially compact), there exists a subsequence $x_{n_k} \to x \in K$. By continuity, $y_{n_k} = f(x_{n_k}) \to f(x) \in f(K)$. Hence $f(K)$ is sequentially compact, therefore compact. □

Extreme Value Theorem and Intermediate Value Theorem

Theorem 6.2 (Extreme Value Theorem (EVT))**.** *If $f\colon [a,b] \to \mathbb{R}$ is continuous, then f attains its maximum and minimum: there exist $x_M, x_m \in [a,b]$ with*

$$f(x_m) \le f(x) \le f(x_M) \quad \text{for all } x \in [a,b].$$

Theorem 6.3 (Intermediate Value Theorem (IVT))**.** *If $f\colon [a,b] \to \mathbb{R}$ is continuous and L is any value between $f(a)$ and $f(b)$ (i.e., $f(a) \le L \le f(b)$ or $f(b) \le L \le f(a)$), then there exists $c \in [a,b]$ with $f(c) = L$.*

Proof of the EVT. By Theorem 6.1, $f([a,b])$ is compact, hence closed and bounded (Heine–Borel, Corollary 4.7). Since $f([a,b])$ is bounded, $M = \sup f([a,b])$ exists. Since $f([a,b])$ is closed and $M \in \mathrm{cl}(f([a,b]))$ (as the supremum of a bounded set is in its closure), we have $M \in f([a,b])$. Hence there exists $x_M \in [a,b]$ with $f(x_M) = M$. The argument for the minimum is analogous, using $m = \inf f([a,b])$. □

Proof of the IVT. Assume without loss of generality that $f(a) < L < f(b)$ (if $L = f(a)$ or $L = f(b)$, take $c = a$ or $c = b$). Define $S = \{x \in [a,b] : f(x) < L\}$. Then S is nonempty (since $a \in S$) and bounded above (by b). Let $c = \sup S$.

Since $f(a) < L$ and f is continuous at a, we have $f(x) < L$ for x near a, so $c > a$. Similarly, $f(b) > L$ and continuity at b give $c < b$. Hence $c \in (a,b)$.

Since $c = \sup S$, there exists a sequence (x_n) in S with $x_n \to c$ (by the ε-characterization of the supremum, Lesson 1, Remark 3.6). Since $f(x_n) < L$ and f is continuous, $f(c) = \lim f(x_n) \le L$.

If $f(c) < L$, then by continuity there exists $\delta > 0$ with $f(x) < L$ for $x \in (c, c+\delta) \cap [a,b]$, contradicting $c = \sup S$. Hence $f(c) = L$. □

EVT in optimization and the IVT in root-finding

The extreme value theorem guarantees that continuous loss functions attain their minima on compact feasible sets. This is the theoretical foundation for constrained optimization: when we write $\theta^* = \arg\min_{\theta \in \Theta} L(\theta)$ with Θ compact and L continuous, the EVT ensures θ^* exists. The intermediate value theorem provides fixed-point existence: if $g\colon [0,1] \to [0,1]$ is continuous, then $h(x) = g(x) - x$ satisfies $h(0) = g(0) \ge 0$ and $h(1) = g(1) - 1 \le 0$, so the IVT gives c with $h(c) = 0$, i.e., $g(c) = c$. This one-dimensional Brouwer fixed-point theorem underlies existence proofs for Nash equilibria.

Example 6.4. Let $g(x) = (e^x - 1)/(e-1)$ on $[0,1]$. Then $g(0) = 0$ and $g(1) = 1$, and g is continuous. The function $h(x) = g(x) - x$ satisfies $h(0) = 0$, so $x = 0$ is a fixed point. To find another, note $g'(0) = 1/(e-1) < 1$, so $g(x) < x$ for small positive x, while $g(1) = 1$. The IVT applied to h on a suitable subinterval confirms $x = 0$ is the unique fixed point.

7 Uniform Continuity

Definition 7.1 (Uniform Continuity)**.** A function $f\colon A \to \mathbb{R}$ is *uniformly continuous on* A if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $x, y \in A$ and $|x - y| < \delta$.

Remark 7.2. The key distinction: in ordinary continuity, δ may depend on both ε and the point c; in uniform continuity, δ depends only on ε and works for *all* pairs of points simultaneously.

Example 7.3. The function $f(x) = x^2$ is continuous on $\mathbb{R}$ but not uniformly continuous on $\mathbb{R}$. Indeed, for any $\delta > 0$, choose $x = 1/\delta$ and $y = x + \delta/2$. Then $|x - y| = \delta/2 < \delta$ but $|f(x) - f(y)| = |y^2 - x^2| = |y - x||y + x| = (\delta/2)(2/\delta + \delta/2) \geq 1$, so taking $\varepsilon = 1$ shows uniform continuity fails.

Example 7.4. If f is Lipschitz on A, i.e., there exists $L > 0$ with $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in A$, then f is uniformly continuous: given $\varepsilon > 0$, take $\delta = \varepsilon/L$. For instance, $f(x) = \sin(x)$ satisfies $|\sin(x) - \sin(y)| \leq |x - y|$ (Lipschitz constant 1), hence is uniformly continuous on $\mathbb{R}$.

Theorem 7.5 (Compact Domain Implies Uniform Continuity)**.** *If $f\colon K \to \mathbb{R}$ is continuous and $K \subseteq \mathbb{R}$ is compact, then f is uniformly continuous on K.*

Proof. Suppose for contradiction that f is continuous on K but not uniformly continuous. Then there exists $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}$, there exist $x_n, y_n \in K$ with $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \geq \varepsilon_0$.

Since K is compact (hence sequentially compact), (x_n) has a subsequence $x_{n_k} \to c \in K$. Since $|x_{n_k} - y_{n_k}| < 1/n_k \to 0$, we also have $y_{n_k} \to c$.

By continuity of f at c: $f(x_{n_k}) \to f(c)$ and $f(y_{n_k}) \to f(c)$. Hence $|f(x_{n_k}) - f(y_{n_k})| \to 0$, contradicting $|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon_0 > 0$. □

Example 7.6. The function $f(x) = x^2$ is uniformly continuous on $[0, 10]$ (by Theorem 7.5, since $[0, 10]$ is compact), even though it fails to be uniformly continuous on all of $\mathbb{R}$ (Example 7.3).

Uniform continuity in value function approximation

In RL with continuous state spaces, value functions $V^\pi(s)$ are often shown to be Lipschitz or uniformly continuous on compact state spaces. Uniform continuity guarantees that if we approximate V^π on a finite grid of states with spacing δ, the approximation error is bounded by ε uniformly—there are no "bad" states where the error blows up. This is essential for the correctness of discretization-based solvers and for bounding the approximation error of neural network function approximators on bounded domains.

8 Exercises

Exercise 8.1 (Classifying sets)**.** $(*)$
For each of the following subsets of $\mathbb{R}$, determine whether it is open, closed, both, or neither. Find its interior, closure, and boundary.

(a) $A = [0, 1)$
(b) $B = \mathbb{Z}$
(c) $C = \{1/n : n \in \mathbb{N}\}$
(d) $D = (0, 1) \cup (1, 2)$

Exercise 8.2 (Compactness verification)**.** $(*)$
Determine which of the following sets are compact, and justify your answers using the Heine–Borel theorem (Corollary 4.7):

(a) $\{x \in \mathbb{R} : x^2 \leq 4\}$

(b) $\mathbb{Q} \cap [0,1]$
(c) $\{1 - 1/n : n \in \mathbb{N}\} \cup \{1\}$
(d) $(0,1]$

Exercise 8.3 (Continuity proof from the definition). $(*)$
Prove directly from Definition 5.1 (the ε-δ definition) that $f(x) = 3x + 7$ is continuous at every $c \in \mathbb{R}$.

Exercise 8.4 (IVT application in RL). $(**)$
Let $V\colon [0,1] \to \mathbb{R}$ be a continuous function with $V(0) = -1$ and $V(1) = 5$. Let $\lambda \in (-1,5)$ be a threshold value.
(a) Use the IVT to prove there exists $s^* \in (0,1)$ with $V(s^*) = \lambda$.
(b) If V is additionally assumed to be strictly increasing, prove that s^* is unique.
Hint: For uniqueness, assume two distinct roots and derive a contradiction with strict monotonicity.

Exercise 8.5 (Uniform continuity failure). $(**)$
Prove that $f(x) = 1/x$ is continuous but not uniformly continuous on $(0,1)$. Then explain why this does not contradict Theorem 7.5.

Exercise 8.6 (Compactness from first principles). $(***)$
Prove directly from the open-cover definition (without using Heine–Borel) that $[0,1]$ is compact.
Hint: Let $\{O_\alpha\}$ be an open cover of $[0,1]$. Define $S = \{x \in [0,1] : [0,x] \text{ is covered by finitely many } O_\alpha\}$. Show S is nonempty, $\sup S = 1$, and $1 \in S$.

9 Summary

- A set $O \subseteq \mathbb{R}$ is *open* if every point has an ε-neighborhood contained in O; a set is *closed* if its complement is open, equivalently if it contains all limits of convergent sequences from the set.
- Arbitrary unions of open sets are open; finite intersections of open sets are open. The dual statements hold for closed sets.
- The *closure* $\mathrm{cl}(A)$ consists of all limits of sequences in A; the *interior* $\mathrm{int}(A)$ is the largest open subset of A; the *boundary* $\partial A = \mathrm{cl}(A) \setminus \mathrm{int}(A)$.
- A set $K \subseteq \mathbb{R}$ is *compact* if and only if it is closed and bounded (*Heine–Borel theorem*), which is equivalent to sequential compactness: every sequence in K has a subsequence converging to a point in K.
- Continuity at c means $|f(x) - f(c)| < \varepsilon$ for $|x - c| < \delta$; equivalently, $x_n \to c$ implies $f(x_n) \to f(c)$. Continuous functions preserve sums, products, quotients, and compositions.
- The *extreme value theorem*: a continuous function on a compact set attains its maximum and minimum.
- The *intermediate value theorem*: a continuous function on $[a,b]$ takes every value between $f(a)$ and $f(b)$, which gives existence of roots and fixed points.
- A function is *uniformly continuous* if δ depends only on ε, not on the point. Every continuous function on a compact set is uniformly continuous.

10 Further Reading

- **Lesson 3** (Differentiation and Integration) builds on continuity to develop differentiability, the mean value theorem, Taylor's theorem, and Riemann integration. The extreme value theorem proved here is used in the proof of Rolle's theorem, and uniform continuity is used to prove that continuous functions are Riemann integrable.
- In Phase 01 (Functional Analysis, Lesson 5), the concepts of open sets, closed sets, and compactness are generalized from $\mathbb{R}$ to arbitrary metric spaces and then to normed spaces. The Heine–Borel characterization fails in infinite-dimensional spaces, making compactness significantly harder to verify.
- The Arzelà–Ascoli theorem, which characterizes compact subsets of $C([a, b])$ (the space of continuous functions), extends the ideas of this lesson to function spaces and is treated in Phase 01.
- For topology in $\mathbb{R}^n$ and general metric spaces, see Rudin [1976, Chapter 2]. For a more abstract topological perspective, see Folland [1999, Chapter 4].
- Abbott [2015, Chapters 3–4] offers an exceptionally clear treatment of compactness and the extreme value theorem, with illuminating discussion of why each hypothesis is needed.
- Students proceeding to measure theory will find that open and closed sets generate the Borel σ-algebra $\mathcal{B}(\mathbb{R})$, which is the natural domain for probability measures on $\mathbb{R}$.

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